

LESSON 2-8 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Finding the Mean**

Maryland Standard 6.SP.B.5c • Learning target – I can find the mean of a data set by sharing the total equally among the values.

REMEMBER

- The **mean** is a fair share: add all the values, then divide by **how many** values there are.
- **Mean = sum ÷ number of values.** Count every value, including repeats and zeros.
- The mean is between the least and greatest values. It does not have to be in the data set.

WORKED EXAMPLE

Kiara read 12, 9, 15, 8, and 11 pages on five nights. Find the mean.

Step 1 Add: $12 + 9 + 15 + 8 + 11 = 55$

Step 2 Count: there are 5 values.

Step 3 Divide: $55 \div 5 = 11$. Check: 11 is between 8 and 15.

Answer 11 pages

WORD BANK

mean – the fair share: add, then divide

data set – all the numbers you collected

sum – the total when you add

PRACTICE

show your work • label units

1. Find the mean of each data set.

a. 6, 9, 12 _____

b. 15, 20, 25, 40 _____

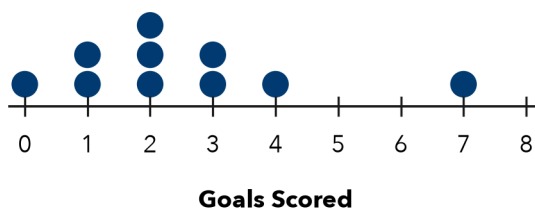
c. 7, 0, 5, 8 _____

d. 3, 3, 9, 4, 6 _____

e. 2.5, 3.5, 4.5 _____

f. 18, 22, 14, 26,
30 _____

2. **Soccer Goals.** The **dot plot** shows the **goals** a team scored in each game. Each dot is one game.



a. Total goals: _____

b. Number of games: _____

c. Mean goals per game: _____

3. **Rainfall.** Rain fell on **five** days: 0.4 in, 1.2 in, **0 in**, 0.9 in and 0.5 in. Add all five, then divide by **5**.

Show your work.

Answer _____ in

- 4. New Value.** The mean of 8, 10, and 12 is 10. A fourth value, 20, is added. What happens to the mean?

☐ It goes up. ☐ It goes down. ☐ It stays the same.

New mean _____

- 5. Find the Error.** To find the mean of 8, 0, 6, and 10, Darnell added to get 24. Then he divided by 3 because “zero does not count.” Explain his error and find the mean.

Frame Darnell is wrong because zero _____. The mean is _____.

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____

AT HOME Ask 4 people at home how many glasses of water they drink in a day. Find the mean.

LESSON 2-8 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Reaching a Target Mean**

Maryland Standard 6.SP.B.5c • Learning target – I can find the value a data set still needs in order to reach a target mean.

REMEMBER

- **Target total = target mean × number of values** (count the missing value too).
- **Missing value = target total – sum so far.** Check: the new total ÷ count should equal the target mean.

WORKED EXAMPLE

Jamal's quiz scores are 78, 85, and 90. What score on quiz 4 gives a mean of 85?

Step 1 Target total: $85 \times 4 = 340$ **Step 2** Sum so far: $78 + 85 + 90 = 253$ **Step 3** Missing: $340 - 253 = 87$. Check: $340 \div 4 = 85$ **Answer 87 points****WORD BANK****target mean** – the mean you want**missing value** – target total – sum so far**target total** – target mean × number of values**PRACTICE***show your work • label units*

1. Complete the table.

Step 1 Target total = target mean × number of values.**Step 2** Missing value = target total – sum so far.

	Values	Target mean	Target total	Sum so far	Missing value
A	3	12		25	
B	4		64	49	

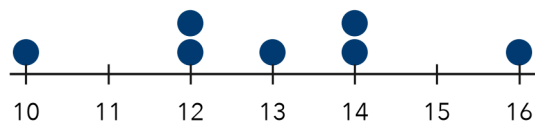
- 2.
- Bowling.**
- Aaliyah bowled
- 112, 98, 105 and 121**
- . What score in
- game 5**
- gives her a mean of
- 110**
- ?

Step 1 Find the target total: 110×5 .**Step 2** Add her four scores.**Step 3** Subtract to find game 5.**Answer** _____ points

- 3.
- Car Wash.**
- A club raised \$45, \$60, \$52, and \$38 on four Saturdays. What must it raise on the fifth Saturday for a mean of \$50?

Step 1 Find the target total: 50×5 .**Step 2** Add the four Saturdays.**Step 3** Subtract.**Answer** _____

4. **Push-Ups.** The dot plot shows how many push-ups Nia did on each of 7 days.



- a. Mean for the 7 days: _____
- b. Day 8 for a mean of 14: _____

5. **Find the Error.** Brianna scored 82, 86, and 78. For a mean of 85 on 4 tests, she wrote $85 \times 3 - 246 = 9$. Explain her error. What score does she need?

Frame Brianna should multiply by ____ because _____. She needs a score of _____.

TEST PRACTICE

choose the best answer

6. Four numbers have a mean of 9. Three of them are 7, 10, and 8. What is the fourth number?
- A** 9 **B** 11 **C** 2 **D** 36
7. A player's points in 5 games have a mean of 14. What is her total number of points?
- A** 2.8 **B** 19 **C** 70 **D** 9

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____